Counting Particles (template)

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How do we count things too small to see?

Targets

- I can say what a mole is.
- I can turn grams into moles with molar mass.
- I can turn moles into a count of particles.

Concepts

- **Mole** (1 mol = 6.02×10^{23}): A counting unit, like a dozen. One mole is 6.02×10^{23} particles.
- **Molar mass** (n = m / M): Grams in one mole. Add the atomic masses in the formula.

What the class predicted, and what is true

- One mole of water and one mole of iron. Which holds more particles? **The same number.** A mole is a count. One mole of anything is 6.02×10^{23} particles.
- Same two samples: one mole of water and one mole of iron. Which has more mass? **Iron.** Same count, but each iron atom is heavier than a water molecule (56 g/mol vs 18 g/mol).

Huddle notes (answers filled in)

Definitions

- **Mole:** The amount that holds **6.02×10^{23}** particles.
- **Molar mass:** The **mass** of one mole, in grams per mole.

Worked example

How many moles are in 36 g of water? (M = 18 g/mol)

- n = m / M
- n = 36 / 18 = **2** mol
- Particles: $2 \times 6.02 \times 10^{23} =$ **12.04×10^{23}**

Label the water molecule

1. The **oxygen** atom.
2. A **hydrogen** atom.
3. The **bond** that holds them.

One worked sample of each skill

Grams to moles

Level 1. A sample of water has a mass of 54 g. Water has a molar mass of 18 g/mol. How many moles is that?
Answer: 3 mol n = m / M = 54 / 18 = 3 mol.

Level 2. A sample of carbon has a mass of 36 g. Its molar mass is 12 g/mol. How many moles is that?
Answer: 3 mol n = 36 / 12 = 3 mol.

Level 3. A sample of CH? has a mass of 4 g. The molar mass of CH? is 16 g/mol. How many moles is the sample?
Answer: 0.25 mol n = 4 / 16 = 0.25 mol.

Level 4. You weigh out 116 g of a compound with M = 58 g/mol. How many moles do you have? Answer in mol.
Answer: 2 mol n = 116 / 58 = 2 mol.

Moles to particles

Level 1. How many particles are in 2 mol? Give the answer in units of 10^{23} (so 6.02×10^{23} is written 6.02).
Answer: 12.04×10^{23} $2 \times 6.02 = 12.04$, so 12.04×10^{23} particles.

Level 2. How many particles are in 0.5 mol? Answer in units of 10^{23}.
Answer: 3.01×10^{23} $0.5 \times 6.02 = 3.01 \times 10^{23}$.

Level 3. How many particles are in 4 mol? Answer in units of 10^{23}.
Answer: 24.08×10^{23} $4 \times 6.02 = 24.08 \times 10^{23}$.

Level 4. How many water molecules are in 54 g of water (M = 18 g/mol)? Answer in units of 10^{23}.
Answer: 18.06×10^{23} n = 54 / 18 = 3 mol. Particles = $3 \times 6.02 = 18.06 \times 10^{23}$.